

Technology Stack

Date	05 July 2026
Team ID	SWTID-2026-4324
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js, HTML5, CSS3, Bootstrap	Creates a responsive, user-friendly interface that works on mobile and desktop.

Technology Stack Details

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask)	Handles business logic, API development, and communication with the database.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	Stores farmer details, crop data, weather information, and recommendations securely.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render (or PythonAnywhere), Docker	Provides easy deployment, scalability, reliable hosting, and cost-effective cloud services for the application.

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub, GitHub Actions	Enables version control, team collaboration, code backup, and automated build and deployment processes.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenWeatherMap API, Google Maps API	OpenWeatherMap API provides real-time weather data (temperature, rainfall, humidity) for accurate crop recommendations. Google Maps API helps identify farm locations and display location-based agricultural information.